

Design for Learnability: Challenging the State-of-the-Art in UX to foster Inclusion and Participation

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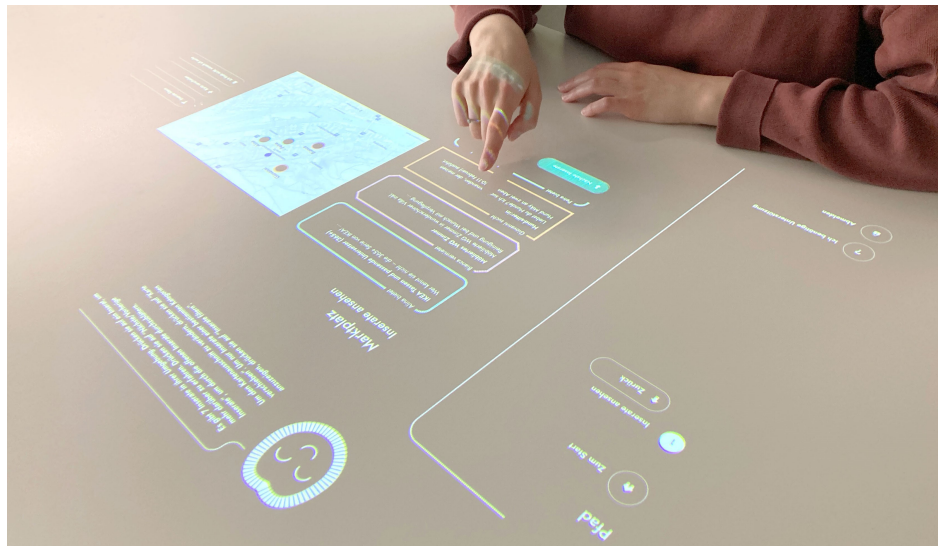


Figure 1: Hybrid Anchor Point based on Interactive Tabletop Projection (Live Paper/Lusee) with one11 App, designed for learnability for simplified access to hybrid community through support functionalities and multi-user interaction in local community centres

ABSTRACT

Design for Intuitive Use is based upon knowledge from previous experiences. People with low digital literacy lack this knowledge, causing difficulties and exclusion from the digital society. In this research project, we focus on *Design for Learnability* as a new paradigm that challenges the state of the art in UX Design to foster inclusion and participation.

To empower this group, so called *Hybrid Anchor Points* are established in local communities. We use an Interactive Tabletop Projection (ITP), which facilitates access to simplify participation. Further it allows for collaborative exploration and learning, taking into account the needs of people with low digital literacy, focusing

on first-use learnability, during-use support and low cognitive load. A pilot study showed a significant reduction in user errors and help interventions.

This research shows that design for learnability, combined with an ITP, can simplify access to the digital society and further advance inclusion and participation.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing devices*; **Collaborative interaction**; **Mixed / augmented reality**; **User studies**; *User centered design*.

KEYWORDS

Interactive Tabletop Projection, Digital Divide, Inclusion, Learnability, Digital Social Innovation, Micro Teachings

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1 INTRODUCTION

To fully participate in today's society, being able to use ICT and keeping up with the rapidly evolving technology is essential. While for the larger part of the population a design for intuitive use is sufficient, there is a significant group who struggle with the basic interactions and concepts [2]. Therefore, design for intuitive use does not feel intuitive. People with low digital literacy, who make up around 20% of the population in Switzerland [1], are inadvertently excluded from full and independent participation in the digital society. For this subgroup, design for intuitive use is not enough and a holistic approach through digital social innovations [6] is required to mitigate the growing digital divide.

The research project *Involvement. Inclusion. Participation. one11* – a self-learning platform enables a new form of life through smart community building has the vision to create a connected solution which enables local, self-sustaining communities to foster inclusion and participation of everyone to lead an independent and fulfilling life. The project establishes *Hybrid Anchor Points* in the local community centers by setting up an Interactive Tabletop Projection (ITP), called Live Paper/Lusee [5] to simplify access and participation. This is one of the possible solutions to meet the needs of people with low digital literacy and contributes to the reduction of the digital divide.

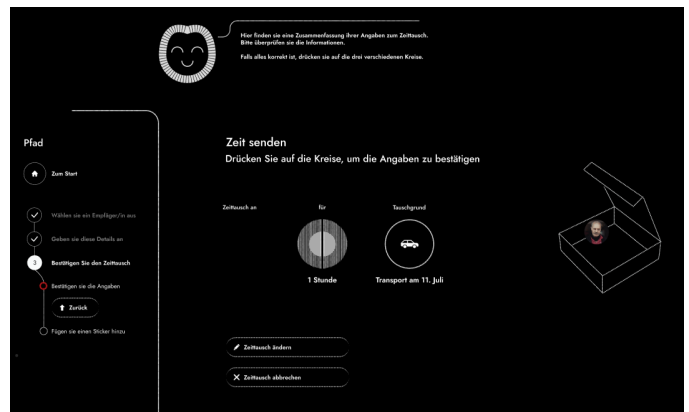
2 UX DESIGN CONSIDERATIONS

One of the main barriers to participation in the digital world is the lack of self-confidence in the use of ICT. Low self-efficacy can also be seen as a vicious circle for people with low digital literacy "They need to gain experience with ICTs to develop their skills, but they are apprehensive about using ICTs because they do not have the needed skills" [2]. Further key steps to lower the barrier of entry are: increasing the desirability of the system by identifying and emphasising motivational factors, and increasing the first-use learnability [10]. For social computing, one of the main motivational factors is the potential connections between individuals. The prospect of new ways in participating in the society can enhance motivation to acquire the necessary skills to interact with ICT [9].

To increase the first-use learnability, one approach is to use micro worlds which are short, interactive learning sessions setup in a simulated environment [8]. This allows safe exploration and learning of functionalities, coupled with hands-on exercises for reinforcement to increase self-efficacy and knowledge [8]. The learnability of the system can be further enhanced by the use of a digital agent that provides instructions during usage in order to reduce stress and anxiety associated with ICT [4]. Reducing the cognitive load can be beneficial for some people with impairments or low digital literacy [3]. This can be achieved through affordances that consider people's knowledge and experience, for example by using metaphors relating to the analogue world for complex concepts or thoughtful language for elements, as modern terms are often abstract [3] [7].



(a) Live Paper/Lusee setup with seamless integration into physical space allowing for multi-user interaction in the community



(b) Screen of application on Live Paper/Lusee which implements several supporting elements: Digital Agent on top of UI, data represented as objects from real-life, step guidance on left side

Figure 2: one11 application on Live Paper/Lusee setup with screen projected on it

3 IMPLEMENTATION

In order to empower and enable people to thrive in the digital society, the research project implements and applies concepts specifically designed to increase desirability, first-use learnability and self-efficacy to the Hybrid Anchor Points. These can be crucial drivers for active participation and reduce the barrier of entry.

To increase the desirability, we aim to create positive experiences when learning how to use the digital platform through the Hybrid Anchor Points. This allows for social interactions before, during and after the use, creating opportunities for members of the community to collaboratively explore the digital content. To enable a multi-user learning experience, the Live Paper / Lusee ITP system is particularly well suited as it allows for a seamless integration into the physical space without any boundaries (Fig. 2a). This creates a blended space, where the technology does not feel intrusive and encourages people to interact with the system while alleviating

fears and apprehensions about the use of ICT systems. The first-use learnability is increased through small, interactive learning sessions. These are developed as micro-worlds that allow users to focus on a specific element or interaction to learn in a controlled and isolated environment [8]. For reinforcement, the learning sessions are accompanied with hands-on exercises, applying Experiential Learning Theory [8]. The exercises relate to real-life knowledge for better comprehension.

Complex processes are broken down into smaller actions, presented through metaphors connecting to the user's everyday life. As shown in Fig. 2b, the user manually places each part of a transaction into a parcel by touching it. When all parts are placed in the parcel, an animated drone picks up the package and delivers it to the recipient.

During use, a digital agent provides written guidance on the interactions in context of the current screen (Fig. 2b). The detailed instructions explain each possible interaction using language related to the user's real-world knowledge. Further guidance is provided by a step display on the left side. It always shows the current step in the process, as well as the previous and future steps, thus providing orientation and clarity. To reduce the cognitive load, the interfaces are simple and reduced in the number of elements.

A first pilot study compares the Hybrid Anchor Points with all its support functionalities to the established web platform on an iPad. Three tasks were performed by 11 participants with low digital literacy as a between-subjects study, leading to 33 data points per technology. Fig. 3 shows the number of errors and help interventions needed on both technologies. The number of help interventions on the Hybrid Anchor Points were reduced significantly ($U=734$, $p=0.012$, $z=-2.51$) with a moderate effect size ($r=0.31$) compared to the execution on the iPad. Similarly, we observed a significant reduction in the number of errors ($U=830$, $p=0.00021$, $z=-3.71$) with a moderate effect size ($r=0.46$).

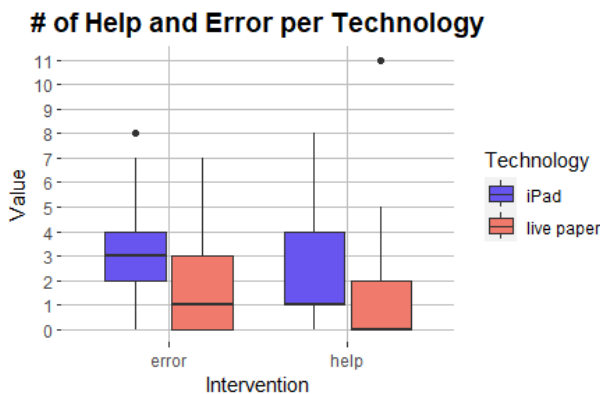


Figure 3: Help and Error rates per Technology during pilot study

4 DISCUSSION

In this research project we challenged the state-of-the-art approach *design for intuitive use* in order to facilitate the inclusion of people with low digital literacy. Our designs diverge from this pattern,

implementing the *design for learnability* paradigm, creating a simplified access and participation in the digital society.

The initial prototype can help to develop the necessary skills to remove the digital barriers and encourage active engagement in the hybrid community. A pilot study shows a significant improvement in error rate and needed help interventions when using the prototype over the standard application on an iPad. Especially the emphasis of first-use learnability through the use of micro worlds shows great promise when designing for people with low digital literacy. In addition, the choice of technology must be carefully considered, as its acceptance is essential for participation. The ITP, with its non-intrusive nature and ability to support multi-user scenarios, can be a powerful choice for this. However, participation in the digital society also exposes people to new threats. Issues such as privacy and security need to be addressed and included in the learning process.

Finally, inclusion is as important as it is complex. Even though we have created a prototype that fosters inclusion, we did not account for all possibilities for which people might be excluded. With this research, while we did not address all issues, we hope to further enhance the digital social innovation and work towards a more inclusive digital society.

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